🧠 INSTRUCTOR-LED COURSE SYLLABUS · PYTHON MACHINE LEARNING

Advanced Machine Learning

An instructor-led course in 13 three-hour sessions. Go beyond supervised models on tables: explain what a model learned, find structure without labels, train neural networks with PyTorch, work with text and recommendations, ship a model behind an API, and finish with a capstone of your own.

Course site: rays-python-machine-learning.netlify.app

`13 sessions × 3 h` `39 class hours` `23 lessons` `2–3 h homework per session` `Python 3.12+`

📚 COURSE DESCRIPTION

This course picks up where Machine Learning Foundations ends. You already know how to frame a problem, split data honestly, build pipelines and tune trees and boosting. Here you learn to explain models and check them for fairness, segment customers and detect anomalies without labels, compress many features into a few, build and train neural networks on an ordinary laptop, classify text and recommend products, and make your work reproducible and deployable.

Every session uses the same company: **Pinecrest Outfitters**, a fictional outdoor-gear retailer, plus two public image datasets (Fashion-MNIST and EuroSAT) for the deep-learning sessions. The data hides real structure to discover: five customer segments, a fraud signal that only shows in combinations, a five-dimensional taste space behind purchases.

🎯 By the end you will be able to

- Explain a model globally and locally with permutation importance, partial dependence, SHAP and LIME, and check it for unequal errors across groups
- Segment customers with K-Means, hierarchical, density-based and mixture models, and choose the number of clusters with evidence
- Detect anomalies and reduce dimensions with PCA, LDA, t-SNE and UMAP without over-reading the pictures
- Build and train neural networks in PyTorch: dense networks, CNNs, LSTMs and transfer learning, on a CPU
- Classify text with TF-IDF models and build recommenders evaluated with hit rate
- Make experiments reproducible with MLflow, serve a model with FastAPI and watch for drift
- Carry a capstone from question to a written model report

Prerequisite: **Machine Learning Foundations** (11 sessions) or equivalent.

✅ PREREQUISITES

- **Machine Learning Foundations** or equivalent experience: scikit-learn pipelines, cross-validation, classification and regression metrics, and tree-based models including gradient boosting.
- Python and pandas at the level of **Python Data Science**.
- A Windows, macOS or Linux laptop with about 3 GB free (no GPU needed). The image datasets and pretrained weights (about 175 MB) download the first time they are used.

💡 FORMAT

13 sessions × 3 hours = 39 class hours, two lessons per session (the studio sessions 12, 13 focus on one). Every lesson on the course site is written for a solo learner and takes about 90 to 120 minutes, so two lessons do not fit whole in one room session. The room protects what is hardest alone: the **mental model**, a **live-coding demo** you type along with, a **guided lab** with help on hand, and **starting** each lesson project. At home you finish the projects and do each lesson's exercises, quiz and journal entry on the free course site: about **two to three hours** between sessions.

Each session runs 0:00 to 3:00: a warm-up on last session, the first lesson, a 10-minute break, the second lesson, and a wrap-up with an exit ticket and the next bring list.

Twice weekly: about 7 weeks. Two homework sets a week, so plan on four to six hours outside class.

Weekly: 13 weeks, with a full week for each homework set.

Intensive days: two sessions a day; homework becomes evening practice and the capstone is finished afterward.

📖 SESSION BY SESSION

Lesson titles link to the lesson pages on the course site. Homework is the time to finish that session's lesson projects, exercises, quizzes and journal entries.

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
1 **Interpreting Models and SHAP** EXPLAINING MODELS	7.1 Interpreting Models 7.2 SHAP Values	Read what a model learned with coefficients, permutation importance, partial dependence and ICE plots, and explain any single prediction as a sum of SHAP contributions.	2–3 h
2 **Local Explanations, Fairness and K-Means** EXPLAINING MODELS CLUSTERING	7.3 LIME and Fairness Checks 8.1 K-Means Clustering	Explain individual predictions with LIME, compare error rates across customer groups, and cluster customers with K-Means, knowing why scaling matters.	2–3 h
3 **Choosing k and Hierarchical Clustering** CLUSTERING	8.2 Choosing k and Customer Segments 8.3 Hierarchical Clustering	Choose the number of clusters with the elbow, silhouette and stability checks, turn clusters into named customer segments, and build and read dendrograms.	2–3 h
4 **Density-Based Clustering and Gaussian Mixtures** CLUSTERING	8.4 Density-Based Clustering 8.5 Gaussian Mixture Models	Find clusters of any shape and flag noise with DBSCAN, OPTICS and HDBSCAN, and fit soft clusters with Gaussian mixtures, choosing components with BIC.	2–3 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
5 **Anomaly Detection and PCA** CLUSTERING DIMENSIONALITY	8.6 Anomaly Detection 9.1 Principal Component Analysis	Detect unusual payments with Isolation Forest and Local Outlier Factor and judge them against known fraud, and compress features with principal component analysis.	2–3 h
6 **LDA, t-SNE and UMAP** DIMENSIONALITY	9.2 Linear Discriminant Analysis 9.3 t-SNE and UMAP	Project data to separate classes with LDA, and make nonlinear maps with t-SNE and UMAP without over-reading them. Includes Checkpoint 1.	2–3 h
7 **Neural Networks and PyTorch** DEEP LEARNING	10.1 Neural Networks from Scratch 10.2 PyTorch Fundamentals	Build a neural network from neurons, loss and gradient descent in NumPy, then write PyTorch tensors, modules, datasets and a training loop.	2–3 h
8 **CNNs and Sequence Models** DEEP LEARNING	10.3 Convolutional Neural Networks 10.4 Sequence Models	Train a convolutional network on Fashion-MNIST and an LSTM that forecasts daily store sales, on a CPU, reading training curves honestly.	2–3 h
9 **Transfer Learning and Text Classification** DEEP LEARNING TEXT & RECS	10.5 Transfer Learning 11.1 Text Classification	Reuse a pretrained torchvision network on satellite images, and build a TF-IDF sentiment model for product reviews.	2–3 h
10 **Recommendations and Reproducibility** TEXT & RECS MLOPS	11.2 Recommendation Systems 12.1 Reproducibility and Experiment Tracking	Recommend products with popularity, item similarity and matrix factorization evaluated by hit rate, and make experiments reproducible with seeds, safe model files and MLflow tracking.	2–3 h
11 **Serving, Monitoring and the Churn Case Study** MLOPS CAPSTONE	12.2 Serving and Monitoring Models 13.1 Case Study: Predicting Churn	Serve a model with FastAPI, watch for data drift, and follow a complete churn project from question to recommendation; choose a capstone problem with a one-paragraph plan.	2–3 h
12 **Capstone Kickoff and Studio I** CAPSTONE	13.2 Capstone Project	Plan a capstone, set an honest split and baselines, and build a first model, with milestone check-ins in the room.	2–3 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
13 **Capstone Studio II and Showcase** CAPSTONE	13.2 Capstone Project	Finish the capstone: error analysis, a model card and a written recommendation (including an honest "do not deploy" when the evidence says so), then present it.	3–5 h

Total out-of-class time: about 27 hours or more across the course.

🕒 SESSION OUTLINES

Times run from the start of each session (0:00 to 3:00). Shift the block to any start time.

SESSION 1 Interpreting Models and SHAP

You build: 7.1 A churn-drivers briefing for Pinecrest's retention team: ranked drivers, grouped importance and the risk curves behind them; **7.2** A call list for the retention team: the 20 riskiest customers, each with its top three reasons in plain words

0:00–0:30	Welcome, environment check and Foundations refresher
0:30–1:30	7.1 Interpreting Models frame and mental model, live demo, guided lab, start the lesson project
1:30–1:40	*Break*
1:40–2:50	7.2 SHAP Values frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 2 Local Explanations, Fairness and K-Means

You build: 7.3 An explanation and fairness audit of the retention campaign: LIME explanations with stability checks, and a by-age-group report with a significance test; **8.1** A first set of Pinecrest customer groups, profiled in dollars and orders, with each group's churn rate

0:00–0:10	Warm-up
0:10–1:20	7.3 LIME and Fairness Checks frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	8.1 K-Means Clustering frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 3 Choosing k and Hierarchical Clustering

You build: **8.2** A segmentation pipeline for Pinecrest customers and a one-page segment brief with sizes, profiles, churn and suggested actions; **8.3** A product family tree: Pinecrest's 200 products grouped by who buys them together, checked against each product's main activity

0:00–0:10	Warm-up
0:10–1:25	8.2 Choosing k and Customer Segments frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	8.3 Hierarchical Clustering frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 4 Density-Based Clustering and Gaussian Mixtures

You build: **8.4** A density map of Pinecrest customers: the dense core segments, their churn rates, and the customers who fit no segment; **8.5** Soft customer segments for Pinecrest: a segment and a confidence for every customer, a list of borderline customers, and a model that scores new customers

0:00–0:10	Warm-up
0:10–1:25	8.4 Density-Based Clustering frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	8.5 Gaussian Mixture Models frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 5 Anomaly Detection and PCA

You build: **8.6** A payment-fraud review queue for June: unsupervised detectors against rules and a supervised model, a budget chosen from costs, and a fairness check on the false alarms; **9.1** Seasonal fingerprints: 96 store-and-product sales histories compressed to one seasonal score each

0:00–0:10	Warm-up
0:10–1:25	8.6 Anomaly Detection frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	9.1 Principal Component Analysis frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 6 LDA, t-SNE and UMAP

You build: 9.2 A loyalty-tier map: the two discriminant axes that separate Basic, Silver and Gold customers, with a tier classifier checked against a baseline; **9.3** A customer map: Pinecrest's behavior segments drawn with t-SNE and UMAP, checked with numbers, with new customers placed on the map

> Checkpoint 1 (30 minutes): a short open-notebook task on Modules 7 to 9.

0:00–0:10	Warm-up
0:10–1:10	`9.2` Linear Discriminant Analysis frame and mental model, live demo, guided lab, start the lesson project
1:10–1:20	*Break*
1:20–2:20	`9.3` t-SNE and UMAP frame and mental model, live demo, guided lab, start the lesson project
2:20–2:50	**Checkpoint 1 (Modules 7 to 9)**
2:50–3:00	Wrap-up

SESSION 7 Neural Networks and PyTorch

You build: 10.1 A churn network written from scratch, compared with a baseline, logistic regression and gradient boosting; **10.2** A PyTorch churn model on every usable feature, with early stopping, saved to disk and compared with logistic regression and gradient boosting

0:00–0:15	**Warm-up and Checkpoint 1 debrief**
0:15–1:20	`10.1` Neural Networks from Scratch frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	`10.2` PyTorch Fundamentals frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 8 CNNs and Sequence Models

You build: 10.3 A clothing classifier with batch normalization and dropout, checkpointed on a validation set, saved to disk and reloaded; **10.4** A calendar-aware LSTM that forecasts next-day sales for four stores, checked against naive forecasts and saved to disk

0:00–0:10	Warm-up
0:10–1:20	10.3 Convolutional Neural Networks frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	10.4 Sequence Models frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 9 Transfer Learning and Text Classification

You build: 10.5 A land-cover classifier: a pretrained ResNet-18 with a new head, trained frozen, then lightly fine-tuned, evaluated class by class and saved; **11.1** A review-triage model that flags unhappy customers for Pinecrest's care team, including the mixed 3-star reviews

0:00–0:10	Warm-up
0:10–1:25	10.5 Transfer Learning frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	11.1 Text Classification frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 10 Recommendations and Reproducibility

You build: 11.2 A "Picked for You" list of five products for every Pinecrest customer, with the model chosen on a validation item and scored once on an untouched test item; **12.1** A tracked, reproducible churn model: four candidates compared in MLflow, the winner saved with skops beside a model card, then reloaded and verified from scratch

0:00–0:10	Warm-up
0:10–1:25	11.2 Recommendation Systems frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	12.1 Reproducibility and Experiment Tracking frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 11 Serving, Monitoring and the Churn Case Study

You build: 12.2 A churn-scoring API with validated input, plus a weekly drift report that tells the care team which weeks to trust; **13.1** A retention campaign plan: a budget-limited offer list, a fairness check and a one-page memo for Pinecrest's customer-care manager

> Capstone kickoff: choose one of the five capstone problems and leave with a written plan.

0:00–0:10	Warm-up
0:10–1:20	12.2 Serving and Monitoring Models frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:40	13.1 Case Study: Predicting Churn frame and mental model, live demo, guided lab, start the lesson project
2:40–2:50	**Capstone choice and one-paragraph plan**
2:50–3:00	Wrap-up

SESSION 12 Capstone Kickoff and Studio I

You build: 13.2 An end-to-end ML project on one of five Pinecrest problems: a notebook that reruns, a saved model, one to three figures and a model card

> Studio: plan, baseline and first model, with milestone check-ins.

0:00–0:10	Warm-up
0:10–1:25	13.2 Capstone Project frame: the brief, the rubric and the plan, plan clinic (Milestone 1), studio I · Milestone 2: load, check, split
1:25–1:35	*Break*
1:35–2:50	13.2 Capstone Project studio I · Milestone 3: baselines and harness, studio I · Milestone 4: a first pipeline, milestone check
2:50–3:00	Wrap-up

SESSION 13

Capstone Studio II and Showcase

You build: 13.2 An end-to-end ML project on one of five Pinecrest problems: a notebook that reruns, a saved model, one to three figures and a model card

> Studio, then a showcase of every learner's capstone.

0:00–0:10	Warm-up
0:10–1:30	13.2 Capstone Project frame: test once, then read the errors, studio II · Milestone 6: test once, interval, errors, studio II · Milestone 7: explain and review
1:30–1:40	*Break*
1:40–2:40	13.2 Capstone Project studio II · Milestone 8: the figure and the card's first sections, showcase prep, showcase
2:40–2:50	Course close and what next
2:50–3:00	Wrap-up

📅 SAMPLE SCHEDULES

No dates: fit them to your calendar. "Twice weekly" puts two sessions in each week (Day 1 and Day 2), finishing in week 7.

SESSION	TWICE WEEKLY (≈7 WEEKS)	WEEKLY (13 WEEKS)
1 Interpreting Models and SHAP	Week 1, Day 1	Week 1
2 Local Explanations, Fairness and K-Means	Week 1, Day 2	Week 2
3 Choosing k and Hierarchical Clustering	Week 2, Day 1	Week 3
4 Density-Based Clustering and Gaussian Mixtures	Week 2, Day 2	Week 4
5 Anomaly Detection and PCA	Week 3, Day 1	Week 5
6 LDA, t-SNE and UMAP	Week 3, Day 2	Week 6
7 Neural Networks and PyTorch	Week 4, Day 1	Week 7
8 CNNs and Sequence Models	Week 4, Day 2	Week 8
9 Transfer Learning and Text Classification	Week 5, Day 1	Week 9
10 Recommendations and Reproducibility	Week 5, Day 2	Week 10
11 Serving, Monitoring and the Churn Case Study	Week 6, Day 1	Week 11
12 Capstone Kickoff and Studio I	Week 6, Day 2	Week 12
13 Capstone Studio II and Showcase	Week 7, Day 1	Week 13

🏁 CHECKPOINT AND PROJECT

- **Self-checks and exit tickets** in every session are practice, not grades: they tell you and your instructor what to revisit.
- **Checkpoint 1 (Session 6, 30 minutes):** an individual, open-notebook task on explaining models, clustering and dimensionality reduction (Modules 7 to 9), scored with a short rubric.
- **Capstone.** In Session 11 you choose one of five Pinecrest problems (demand forecasting, a fraud review queue, review triage, recommendations, or customer segments) and write a one-paragraph plan. Sessions 12 and 13 are studios with milestone check-ins, ending in a showcase. You hand in a notebook that reruns, a model card and a one-page recommendation. An honest "do not deploy" backed by evidence is a full answer.

🧰 REQUIRED SOFTWARE AND DATA

- **Python 3.12 or newer:** the same environment as Machine Learning Foundations
- **The course packages:** `python -m pip install -r requirements.txt` (adds nothing new if you took Foundations): SHAP, LIME, UMAP, PyTorch and torchvision (CPU build), MLflow, FastAPI
- **JupyterLab or VS Code:** as in Foundations
- **The course data:** `pinecrest-ml-data.zip`, plus Fashion-MNIST, EuroSAT and two pretrained networks that PyTorch downloads on first use (about 175 MB)
- **No GPU, accounts or paid tools:** every deep-learning session is sized to run on a laptop processor in minutes

Everything installs once, at home, before Session 1, following Lesson 0.1. Your instructor sends a setup guide and an environment check a week ahead.

✉️ FOR INSTRUCTORS

Want to teach this course? A complete instructor kit for Advanced Machine Learning (facilitator guides, slides, labs, workbooks, answer keys and assessments for all 13 sessions) is available under license. Ask about the instructor kit.

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